

Exercise 1 For the given task schedule, do the following:

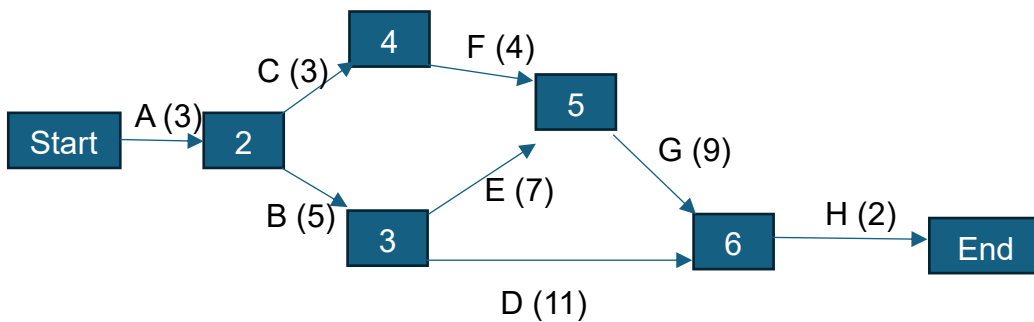
1. Draw the Network Diagrams in AOA and AON format.
2. Find the Critical Paths (critical tasks and critical path length).
3. Calculate the Earliest Start and Latest Start time of each task.
4. Calculate the Total Float and Free Float of each task.

NO.	Task	Predecessor Tasks	Duration
1	A		3
2	B	A	5
3	C	A	3
4	D	B	11
5	E	B	7
6	F	C	4
7	G	E, F	9
8	H	D, G	2

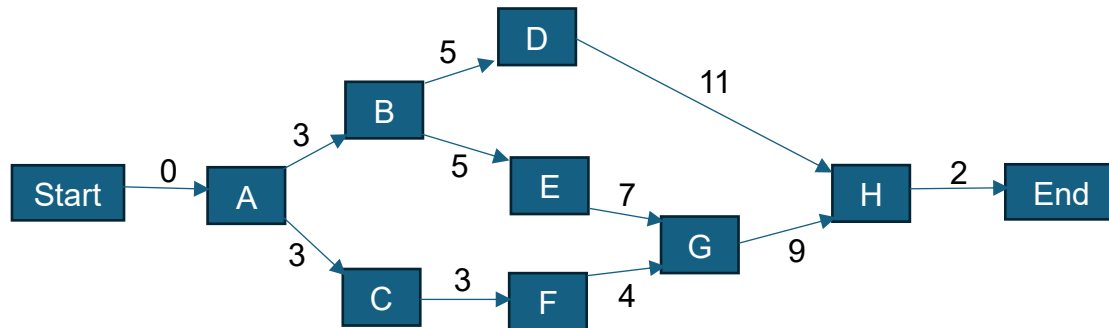
Solution

1. Draw the Network Diagrams in AOA and AON format.

AOA



AON



2. Find the Critical Paths (critical tasks and critical path length).

Path 1:

$$\begin{aligned} &A \rightarrow B \rightarrow D \rightarrow H \\ &= 3 + 5 + 11 + 2 = 21 \end{aligned}$$

Path 2:

$$\begin{aligned} &A \rightarrow B \rightarrow E \rightarrow G \rightarrow H \\ &= 3 + 5 + 7 + 9 + 2 = 26 \end{aligned}$$

Path 3:

$$\begin{aligned} &A \rightarrow C \rightarrow F \rightarrow G \rightarrow H \\ &= 3 + 3 + 4 + 9 + 2 = 21 \end{aligned}$$

⇒ Critical Path: A → B → E → G → H
Critical Tasks: A, B, E, G, H
Critical Path Length: 26 (time units)

3. Calculate the Earliest Start and Latest Start time of each task.

Earliest start: forward

$$t_i = \max_{j \in P(i)} \{t_j + t_{ij}\}$$

Predecessor nodes
(before i)

Duration of the
edge (i, j)

Latest start: backward

$$T_i = \min_{j \in S(i)} \{T_j - t_{ij}\}$$

Successor nodes
(after i)

Duration of the
edge (i, j)

Earliest Star

Task	Predecessors P(i)	Tính toán	t_i (ES)
A	Start	$t_{\text{Start}} = 0$	0
B	A	$t_A + t_{AB} = 0 + 3 = 3$	3
C	A	$t_A + t_{AC} = 0 + 3 = 3$	3
D	B	$t_B + t_{BD} = 3 + 5 = 8$	8
E	B	$t_B + t_{BE} = 3 + 5 = 8$	8
F	C	$t_C + t_{CF} = 3 + 3 = 6$	6
G	E, F	$\max \{t_E + t_{EG}, t_F + t_{FG}\} = \max \{8 + 7, 6 + 4\} = \max \{15, 10\}$	15
H	D, G	$\max \{t_D + t_{DH}, t_G + t_{GH}\} = \max \{8 + 11, 15 + 9\} = \max \{19, 24\}$	24

Latest Start

Task	Successors S(i)	Tính toán	T_i (LS)
H	End	$T_H = 26 - 2 = 24$	24
G	H	$T_H - t_{GH} = 24 - 9 = 15$	15
D	H	$T_H - t_{DH} = 24 - 11 = 13$	13
E	G	$T_G - t_{EG} = 15 - 7 = 8$	8
F	G	$T_G - t_{FG} = 15 - 4 = 11$	11
B	D, E	$\min \{T_D - t_{BD}, T_E - t_{BE}\} = \min \{13 - 5, 8 - 5\} = \min \{8, 3\}$	3
C	F	$T_F - t_{CF} = 11 - 3 = 8$	8
A	B, C	$\min \{T_B - t_{AB}, T_C - t_{AC}\} = \min \{3 - 3, 8 - 3\} = \min \{0, 5\}$	0

Result

Task	t_i (ES)	T_i (LS)
------	------------	------------

A	0	0
B	3	3
C	3	8
D	8	13
E	8	8
F	6	11
G	15	15
H	24	24

4. Calculate the Total Float and Free Float of each task.

Total float:

$$M_i = T_i - t_i$$

Free float:

$$m_i = t_j - t_i - t_{ij}$$

Duration of
current node i
(Length of edge ij)

Earliest start
among the
successor
nodes (after i)

Earliest start
of current
node i

A

$$M_A = 0 - 0 = 0$$

$$m_A = \min(3, 3) - 0 - 3 = 0$$

B

$$M_B = 3 - 3 = 0$$

$$m_B = \min(8, 8) - 3 - 5 = 0$$

C

$$M_C = 8 - 3 = 5$$

$$m_C = 6 - 3 - 3 = 0$$

D

$$M_D = 13 - 8 = 5$$

$$m_D = 24 - 8 - 11 = 5$$

E

$$M_E = 8 - 8 = 0$$

$$m_E = 15 - 8 - 7 = 0$$

F

$$M_F = 11 - 6 = 5$$

$$m_F = 15 - 6 - 4 = 5$$

G

$$M_G = 15 - 15 = 0$$

$$m_G = 24 - 15 - 9 = 0$$

H

$$M_H = 24 - 24 = 0$$

$$m_H = 0 \text{ (task cuối)}$$

Task	Total Float M_i	Free Float m_i
A	0	0
B	0	0
C	5	0
D	5	5
E	0	0
F	5	5
G	0	0
H	0	0